3-Choosability of Toroidal Grids: the Problem, the Solved Cases, and Where We Are

Reference document for Sid

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1 The problem

List colouring. A *list assignment* L gives each vertex v a set $L(v)$ of allowed colours. G is *L-colourable* if there is a proper colouring with $c(v) \in L(v)$ for all v. G is *k-choosable* if it is L-colourable whenever every $|L(v)| \geq k$. The smallest such k is the *choice number* $\chi_\ell(G)$. Always $\chi(G) \leq \chi_\ell(G)$, and the gap can be large: $K_{n,n}$ is 2-colourable but its choice number grows with n. So knowing G is 3-colourable says little about lists.

The graph. The toroidal grid $C_m \square C_n$ has vertices (r, c), $r \in \mathbb{Z}_m$ (row), $c \in \mathbb{Z}_n$ (column). (r, c) is adjacent to $(r \pm 1, c)$ (*column edges*: each column is a copy of C_m) and $(r, c \pm 1)$ (*row edges*: each row is a copy of C_n). It is 4-regular, so greedy colouring gives 5-choosable, and it is 3-colourable for all $m, n \geq 3$.

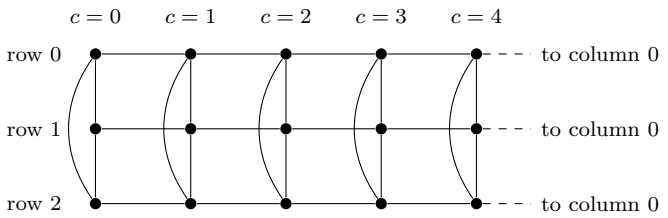


Figure 1: $C_3 \square C_5$: each column is a triangle, each row a 5-cycle (dashed edges wrap around).

Conjecture 1.1 (Cai, Wang, Zhu, 2010)**.** *Every toroidal grid $C_m \square C_n$ $(m, n \geq 3)$ is 3-choosable.*

Split by parity of m, n:

case	status	by
both even	3-choosable	Alon–Tarsi, easy (bipartite)
one odd, one even	3-choosable	Alon–Tarsi, Li–Shao–Petrov–Gordeev 2023
$m = 3$, n odd	3-choosable (claimed)	Galvin for $n = 3$; your partner for all odd n
both odd, $m, n \geq 5$	**open**	this is the target

2 The main tool: Alon–Tarsi

Graph polynomial. Orient each edge uv as $u \to v$ and put $P_G = \prod_{u \to v} (x_u - x_v)$. A colouring c is proper iff $P_G(c) \neq 0$.

Lemma 2.1 (Combinatorial Nullstellensatz)**.** *If $\deg P = \sum_v t_v$ and the coefficient of $\prod_v x_v^{t_v}$ in P is nonzero, then for any sets S_v with $|S_v| > t_v$ there are $s_v \in S_v$ with $P(s) \neq 0$.*

So if $[\prod x_v^{t_v}] P_G \neq 0$, then G is colourable from any lists with $|L(v)| \geq t_v + 1$. The *Alon–Tarsi number* $AT(G)$ is the least k such that some monomial with all $t_v \leq k - 1$ has nonzero coefficient. Always $\chi_\ell \leq AT$.

Key simplification for our graphs. $C_m \square C_n$ has $N = mn$ vertices and $2N$ edges, so $\deg P_G = 2N$. A monomial with every $t_v \leq 2$ and total degree $2N$ must be $\prod_v x_v^2$. Hence

$$AT(C_m \square C_n) \leq 3 \iff [\textstyle\prod_v x_v^2]\, P_G \neq 0.$$

Combinatorially, this coefficient equals $\pm(EE - EO)$ for an orientation with all out-degrees 2, where EE/EO count spanning Eulerian subgraphs with an even/odd number of edges (the Alon–Tarsi theorem).

Case 1: both even

G is bipartite, so every cycle, hence every Eulerian subgraph, has an even number of edges: $EO = 0$ and $EE \geq 1$ (the empty subgraph). Take any Eulerian orientation (all out-degrees 2, exists since G is 4-regular). So $AT \leq 3$, and G is 3-choosable.

Case 2: one odd, one even

Not bipartite, so the counting is no longer trivial. Li–Shao–Petrov–Gordeev computed the coefficient with the *coefficient formula* below and showed it is nonzero. Same conclusion: $AT = 3$.

Lemma 2.2 (Lasoń; Karasev–Petrov)**.** *If* $\deg Q \le \sum t_v$ *and* $|A_v| = t_v + 1$*, then*

$$[\textstyle\prod x_v^{t_v}]\, Q = \sum_{\varphi \in \prod A_v} \dfrac{Q(\varphi)}{\prod_v N_v(\varphi(v))}, \qquad N_v(a) = \textstyle\prod_{b \in A_v, b \neq a} (a - b).$$

Choosing every $A_v = \{1, \omega, \omega^2\}$ (cube roots of unity) turns the coefficient into a sum over *3-colourings with colours* $1, \omega, \omega^2$, weighted by products of differences. On a grid this sum factorises column by column: a *transfer matrix* T indexed by colourings of one column, and the coefficient is $\operatorname{tr}(T^n)$.

Why both odd is different

Li–Shao–Petrov–Gordeev also proved $AT = 4$ when m, n are both odd: the coefficient of $\prod x_v^2$ is exactly 0. Your partner reproved this for $m = 3$ in two lines: the nonzero eigenvalues of T come in pairs $\pm\lambda$ ($\pm i\sqrt{3}$), so $\operatorname{tr}(T^n) = \sum(\lambda^n + (-\lambda)^n) = 0$ for odd n.

Ponder

Why does the $\pm$ pairing of eigenvalues kill the trace exactly for odd n? Which symmetry of the column states (he calls them even/odd states) produces the pairing?

So for odd × odd, Alon–Tarsi cannot work: any proof must be genuinely about lists, not about one polynomial coefficient.

3 Case 3: $m = 3$, n odd

3.1 $n = 3$: Galvin

$C_3 \square C_3 = K_3 \square K_3$ is the line graph of $K_{3,3}$ (cells of a 3×3 grid, adjacent if in the same row or column). Galvin's theorem (the Dinitz conjecture proof) gives 3-choosability. Galvin's method is the *kernel method*, which reappears in Section 5.

3.2 Your partner's route, step by step

His work came in three stages: an Alon–Tarsi result on a damaged graph, a full (partly computer-checked) proof in his paper [P], then a computer-free proof (the October 20 note). The ideas, in the order he found them:

Step 1: what is *almost* true? Delete one edge. For G the coefficient is 0. Delete one edge e: $G - e$ has $2N - 1$ edges, so the natural monomial is $x_w \prod_{v \neq w} x_v^2$ (one vertex w gets exponent 1).

Theorem 3.1 (Partner, "Deleting one edge…")**.** *For odd* n*, every edge* e *and every* w *not on the cycle (column triangle or row cycle) through* e*:* $c_w = [x_w \prod_{v \neq w} x_v^2] P_{G-e} \neq 0$*, with an exact formula (powers of 3).*

How: the coefficient formula with cube roots of unity. In an undamaged column a contributing colouring is a permutation of $(1, \omega, \omega^2)$: 6 *states*. The transfer matrix is $T = \sigma(S^2 - S)$ (S: multiply the state by ω; σ: ± 1 on even/odd permutations), so $T^3 = -3T$ and everything diagonalises in closed form. The damaged column (12 contributing triples) breaks the $\pm$ symmetry, and the factor $\varphi(w)$ at w shifts the eigen-mode; summing gives nonzero powers of 3.

Consequences:

- $G - e$ is 3-choosable, even with $|L(w)| = 2$.
- **If some edge uv of G has $L(u) \cap L(v) = \emptyset$, G is L-colourable** (colour $G - e$; the edge uv is automatically fine).

So a bad list assignment must have overlapping lists on *every* edge. This is the first structural restriction on a counterexample.

Ponder

Why must w be off the cycle of e? (For a row edge, w in that row gives $\beta_0 = \omega^0 - \omega^0 = 0$.) What does that say about which vertex can have the short list?

Step 2: the strip lemma (pinning colours at the ends). Cut the torus open: $H_m = C_3 \square P_m$ (a strip of m columns). An *end multiset* S picks three vertices in the first and last columns with multiplicity ≤ 2; let k be how many lie in the last column and $s = \sum \mu_S(v)\,\mathrm{row}(v) \bmod 3$.

Theorem 3.2 (Odd strip lemma)**.** *For odd $m \geq 3$, $1 \leq k \leq 2$ and $s \not\equiv 0$: $c_S \neq 0$, where $c_S = -\frac{g}{9}\mu_k^{m-1}(\omega^s - \omega^{2s})$. Hence H_m is L-colourable whenever $|L(v)| \geq 3 - \mu_S(v)$.*

Meaning: you may *fix* a colour at a vertex of multiplicity 2 (list of size 1) and *forbid* a colour at a vertex of multiplicity 1 (list of size 2), and the strip is still colourable. This is the tool for gluing.

Ponder

Where exactly does $s \not\equiv 0 \pmod 3$ come from, and why does the formula use $(-1)^m$ so that odd m matters? (For even m the bracket is $\omega^s + \omega^{2s}$.)

Step 3: the prism lemma (exact obstruction for two columns). The prism is two triangles $U = u_0u_1u_2$, $V = v_0v_1v_2$ joined by u_tv_t.

Lemma 3.3 (Prism lemma)**.** *Let $|L(u_0)| \geq 3$ and $|L(w)| \geq 2$ elsewhere. P is not L-colourable iff* ***(Z)*** *$L(v_0) = L(v_1) = L(v_2)$ is one 2-set, or* ***(W)*** *$L(u_1) = L(u_2) = D$, $L(u_0) = D \cup \{\zeta\}$, $L(v_1) = L(v_2) = E$ and $L(v_0) \setminus E = \{\zeta\}$ (D, E 2-sets).*

In (W), u_1, u_2 use up D, forcing $u_0 = \zeta$; v_1, v_2 use up E, forcing $v_0 = \zeta$: conflict. The proof is Hall's theorem on triangles plus case analysis.

Two-column extension (corollary): given fixed colours on the left of column M and on the right of column N, if M, N cannot be filled then $|L_N(r') \cap L_N(r'')| = 2$ and the colour forced at (r', N) is the unique colour of $L_N(r') \setminus L_N(r'')$. So failure is predictable, and can be dodged in advance.

Step 4: the propagation lemma (from [P]). Let $F(c) = L(0,c) \cap L(1,c) \cap L(2,c)$, the colours common to column c. For any list assignment with no colouring (lists of size exactly 3):

$$L(r,c) \setminus F(c) \subseteq L(r, c+1) \qquad \text{for all } r, c. \tag{$*$}$$

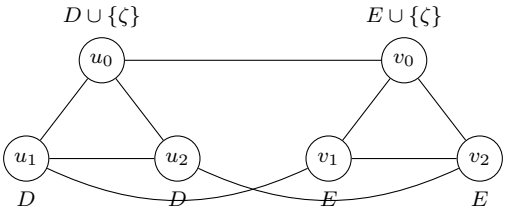


Figure 2: Configuration (W): both triangles are forced to put ζ on the edge u_0v_0.

Status

This is [P, Lemma 3.5]. We have not seen its proof. It rests on further lemmas of [P] and on the column-edge case of the edge-deletion theorem. Everything else in the proof is in the notes we have.

Step 5: the proof. Suppose L (all lists of size 3, $n \geq 5$ odd) has no colouring.

Case A: $F(j) \not\subseteq F(j{+}1)$ for some j. Take $\alpha \in F(j) \setminus F(j{+}1)$ and a row r with $\alpha \notin L(r, j{+}1)$. Name columns $C_0, \ldots, C_3 = j, \ldots, j+3$. Delete C_1, C_2: what remains is a strip of $n-2$ columns (odd) from C_3 around to C_0. Use the strip lemma with $S = \{(r, C_0), (r, C_0), (r', C_3)\}$: fix $(r, C_0) = \alpha$ and forbid a cleverly chosen β at (r', C_3) ($s = 2r + r' \equiv r' - r \not\equiv 0$). Now fill C_1, C_2. Row r is automatically fine since $\alpha \notin L(r, C_1)$. The two-column lemma says the only way to fail is a specific forced colour at (r', C_3), and β was chosen to be exactly that colour. Contradiction.

Case B: $F(j) \subseteq F(j+1)$ for all j. Around the cycle, all $F(j)$ equal one set F. By $(*)$, $L(r, j) \subseteq L(r, j+1)$, and sizes are 3, so every row has a *constant* list A_r. Colour $C_3 \square C_3$ from these lists (Galvin) to get column states X, Y, Z, pairwise compatible. Colour $C_3 \square C_n$'s columns $X, Y, Z, Y, Z, \ldots, Y, Z$ ($1 + 2 \cdot \frac{n-1}{2} = n$ columns). Contradiction.

Ponder

(1) In Case A, why do we need $n \geq 5$? (Columns C_0 and C_3 must not be adjacent.) (2) In Case B, why does $X, Y, Z, Y, Z, \ldots$ need n odd, and why does it close up? (3) The whole proof only uses the structure of *triangles* in Steps 3 and the strip coefficient. Which steps would survive if columns were 5-cycles?

Status

What we checked: the edge-deletion coefficients match our independent computation for $n = 3, 5, 7, 9$; a SAT search found no bad assignment of $C_3 \square C_5$ with 4 colours. Not yet checked: [P] (Lemma 3.5, Theorem 4, and what they use).

4 The open case: m, n odd, ≥ 5

First question: which known methods could even work? Alon–Tarsi is out ($AT = 4$). The other classical method is the kernel method. Testing it was our first contribution.

5 Our work: the kernel method

Kernels. A *kernel* of a digraph is an independent set K such that every vertex outside K has an arc *into* K. D is *kernel-perfect* if every induced subdigraph has a kernel.

Lemma 5.1 (Bondy–Boppana–Siegel)**.** *If D is a kernel-perfect orientation of G and $|L(v)| > d^+(v)$ for all v, then G is L-colourable.*

Idea: for each colour γ, take the vertices whose lists contain γ, colour a kernel of that subdigraph with γ; every other such vertex loses γ but also an out-neighbour, so $|L| > d^+$ is preserved. Induct.

So we want an orientation of $C_m \square C_n$ with all out-degrees ≤ 2 (hence exactly 2, since there are $2N$ edges) that is kernel-perfect. Call these *admissible*.

How we got there. The 2-kernel project trained the right reflexes: test by exhaustive computation first, then explain. First observation (from Theorem B of the 2-kernel work): an admissible orientation is strongly connected, so if it had no odd directed cycle the parity argument would make G bipartite. So odd directed cycles always exist, and kernel-perfectness needs all of them to be non-induced (an induced odd directed cycle has no kernel).

The search. Every admissible orientation, exhaustively:

graph	admissible orientations	kernel-perfect
$C_3 \square C_3$	148	12
$C_3 \square C_5$	2116	0
$C_3 \square C_7$	33028	0
$C_3 \square C_9$	525316	12
$C_5 \square C_5$	143224	0

Surprise: $C_3 \square C_9$ works and $C_3 \square C_5, C_3 \square C_7$ do not. The 12 for $C_3 \square C_9$ are the $C_3 \square C_3$ ones *repeated* three times (lifts along $c \mapsto c \bmod 3$). Lifting in both directions (to $C_9 \square C_9$) fails.

Structure. In a kernel-perfect admissible orientation of $C_3 \square C_n$:

1. each column triangle is transitive (a directed triangle is induced and kernel-less): source S, middle M, sink T;
2. degrees force: both row arcs *into* S, both *out of* T, M one each.

Let $e_r(c) = \pm 1$ be the direction of row r's arc between columns $c, c+1$. At each column exactly one row flips $+ \to -$ and one flips $- \to +$, so the number of $+$'s never changes. Hence at every boundary all rows point the same way except one row p_c, and $p_c \neq p_{c+1}$:

Candidate orientations $\longleftrightarrow$ a direction ε and a proper 3-colouring p of the cycle C_n.
Column c: $T = p_{c-1}$, $S = p_c$, $M =$ the third row (for $\varepsilon = +$).

Call p *Latin* if every three consecutive values are distinct. That forces $p_c = p_0 \pm c$, hence $3 \mid n$.

Theorem 5.2 (only if, hand proof)**.** *If $n \geq 5$ is odd and p is not Latin, the orientation contains an induced odd directed cycle.*

How: a *winding cycle* goes once around the torus, moving right along rows and sometimes stepping inside a column ($S \to M \to T$).

- **Rule A**: always ride the middle vertex (b_c = third row of p_c, p_{c+1}). It is induced, and it steps once per *rainbow* triple (p_{c-1}, p_c, p_{c+1}). The number of non-rainbow ("aba") triples is even (sign changes of $p_{c+1} - p_c$), so the number of rainbow ones is odd and the length $n + R$ is *even*. Useless by itself.
- **Gadget**: where a rainbow triple is followed by an aba one (p_c, p_{c+1}, p_{c+2} distinct, $p_{c+3} = p_{c+1}$), change one choice: $b_c := p_{c+1}$. This adds or removes exactly one step, keeps the cycle induced, and makes the length odd.

Non-Latin p always has such a spot, since it has both triple types.

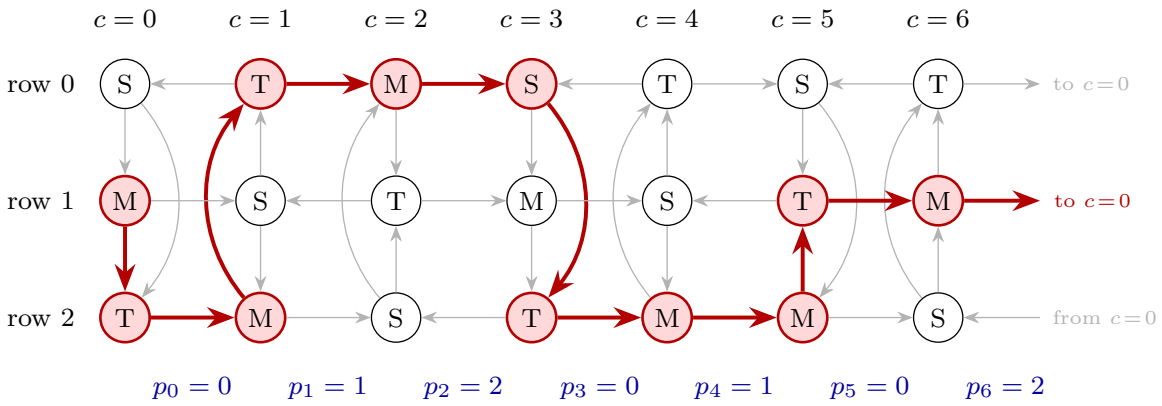


Figure 3: $C_3 \square C_7$, $p = (0, 1, 2, 0, 1, 0, 2)$: the odd induced directed cycle (length 11) built by Rule A plus the gadget at $c = 2$.

Theorem 5.3 (if, computer-assisted)**.** *For odd k, the 12 Latin orientations of $C_3 \square C_{3k}$ are kernel-perfect.*

How: kernel existence is checked column by column by a 20-state automaton. Every induced subdigraph of $C_3 \square C_{3k}$ is a product of k boolean "block" matrices, and has a kernel iff the product has a nonzero trace. The set of all products is finite, so one closure computation covers every k. It fails for even k (e.g. $C_3 \square C_6$).

Result. For odd n: $C_3 \square C_n$ has a kernel-perfect admissible orientation iff $3 \mid n$, then exactly 12. And $C_5 \square C_5$ has none.

Ponder

(1) Why period 3? (Order-3 Latin squares.) Is there a hand proof that Latin lifts are kernel-perfect, e.g. via Galvin's stable-matching argument? (2) Why does it fail for even k? (3) For $m \geq 5$ the columns are 5-cycles, which can be oriented in more ways. Can the encoding lemma be generalised to prove that no $C_m \square C_n$ with $m, n \geq 5$ works?

What this buys the paper: both classical methods (Alon–Tarsi and kernels) provably fail on most of the open case. That is the justification for a new method like your partner's.

6 Our work: first steps toward $m = 5$

Edge deletion still works. Every coefficient c_w (w off the cycle of e) is nonzero for $C_5\square C_5$, $C_5\square C_7$, $C_5\square C_9$, $C_5\square C_{11}$, $C_7\square C_5$, $C_7\square C_7$. For $C_5\square C_5$ and $C_5\square C_7$ this is exact (computed modulo a prime), so for these the "edge with disjoint lists" corollary is proved.

But the algebra gets harder. Nonzero eigenvalues of the transfer matrix:

m	column states	nonzero eigenvalues
3	6	$\pm i\sqrt{3}$
5	30	6 pairs $\pm i\lambda$
7	126	25 pairs $\pm i\lambda$

A closed form like $T^3 = -3T$ is hopeless for $m = 5$. A computer-assisted proof for each fixed m is realistic: as $n \to \infty$ the largest pair dominates, so $c_w \neq 0$ for $n \geq n_0$ once its amplitude is nonzero; check $n < n_0$ directly.

Counterexample search. A SAT loop proposes lists and refutes them with colourings. $C_3\square C_5$, 4 colours: no bad assignment. $C_5\square C_5$: unfinished, too slow as written.

Ponder

Case B of the $m = 3$ proof needs a colouring of $C_3\square C_3$ from row-constant lists. For $C_5\square C_n$, what replaces it? (Hint: $C_5\square C_3 \cong C_3\square C_5$, which your partner's theorem covers. Repeat its three columns as $X, Y, Z, Y, Z, \ldots$.)

7 What remains, in order

These are the research steps; everything else is writing. "Done" criteria are in italics.

A. Make the $m = 3$ theorem airtight (required).

1. Get [P]. Verify Lemma 3.5 (propagation) line by line, plus everything it uses. *Done when we can prove* $(*)$ *ourselves.*
2. Verify the strip lemma computation ([P, Theorem 4], Steps 1–3). *Done when our own code reproduces* c_S *for small* m *and the derivation checks.*
3. Test $(*)$ and the full theorem on small cases by computer: exhaustive search for bad assignments of $C_3\square C_5$, $C_3\square C_7$ with 5–6 colours, using the overlap restriction. *Done when no counterexample appears in feasible ranges.*

B. Finish the kernel section (required, ours).

1. Learn both proofs in Section 5 well enough to present without notes.
2. Optional but valuable: a hand proof that Latin lifts are kernel-perfect for odd k.

C. Extend to $m = 5$ (what makes it a strong paper). For $C_5 \square C_n$, all odd n:

1. Edge deletion for all odd n (dominant eigenvalue + finite check).
2. Odd strip lemma for $C_5 \square P_k$: define end multisets, compute c_S, find the analogue of $s \not\equiv 0$, prove nonvanishing.
3. Two-column extension for 5-cycle columns (replace the prism lemma). Odd cycles with 2-lists are colourable unless all lists are equal: the starting point.
4. Propagation ($*$) for $m = 5$ (depends on what [P]'s proof uses).
5. Case B via $C_3 \square C_5$ (see the last Ponder box).
6. Assemble Cases A and B. *Done when $C_5 \square C_n$ is 3-choosable for all odd n.*

In parallel: a faster counterexample search for $C_5 \square C_5$ (symmetry breaking, better solver). If one exists, the conjecture is false, which is an even bigger result.

D. General m (stretch). Once $m = 5$ works, see which lemmas are uniform in m.

When is it enough for a good journal?

- **Minimum:** A + B. A verified proof for all $C_3 \square C_n$ (an infinite family of an open conjecture), plus a theorem explaining why the standard methods fail. Solid for a combinatorics journal.
- **Strong:** A + B + C. Two infinite families, and a method shown to generalise.
- **Exceptional:** D, or a counterexample.